

Bayesian Estimation

As mentioned in Section 3.1, a comparison of two estimators from their risk functions will be inconclusive whenever the graphs of these functions cross. This difficulty will not arise if the performance of an estimator is measured with a single number. In a Bayesian approach to inference the performance of an estimator δ is judged by a weighted average of the risk function, specifically by

$$\int R(\theta, \delta) d\Lambda(\theta), \quad (7.1)$$

where Λ is a specified probability measure on the parameter space Ω .

7.1 Bayesian Models and the Main Result

The weighted average (7.1) arises as expected loss using δ in a *Bayesian probability model* in which both the unknown parameter and data are viewed as random. For notation, Θ is the random parameter with θ a possible value for Θ . In the Bayesian model,

$$\Theta \sim \Lambda,$$

with Λ called the *prior* distribution because it represents probabilities before data are observed, and P_θ is the conditional distribution of X given $\Theta = \theta$, that is,

$$X|\Theta = \theta \sim P_\theta.$$

Then

$$E[L(\Theta, \delta(X)) \mid \Theta = \theta] = \int L(\theta, \delta(x)) dP_\theta(x) = R(\theta, \delta),$$

and by smoothing,

$$EL(\Theta, \delta) = EE[L(\Theta, \delta) \mid \Theta] = ER(\Theta, \delta) = \int R(\theta, \delta) d\Lambda(\theta).$$

In Bayesian estimation, the choice of the prior distribution Λ is critical. In some situations Θ may be random in the usual frequentist sense with a random process producing the current parameter Θ and other random parameters in the past and future. Then Λ would be selected from prior experience with the random process. For instance, the parameter Θ may be the zip code on a letter estimated using pixel data from an automatic scanner. The prior distribution here should just reflect chances for various zip codes. More commonly, the parameter Θ cannot be viewed as random in a frequentist sense. The general view in these cases would be that the prior Λ should reflect the researchers' informed subjective opinion about chances for various values of Θ . Both of these ideas regarding selection of Λ may need to be tempered with a bit of pragmatism. Calculations necessary to compute estimators may be much easier if the prior distribution has a convenient form.

An estimator that minimizes (7.1) is called *Bayes*. Lacking information from data X , the best estimate is just the constant minimizing $EL(\Theta, d)$ over allowed values of d . The following result shows that a Bayes estimator can be found in a similar fashion. The key difference is that one should now minimize the conditional expected loss given the data, that is $E[L(\Theta, d) \mid X = x]$. This conditional expected loss is called the *posterior risk* and would be computed integrating against the conditional distribution for Θ given $X = x$, called the *posterior distribution* of Θ .

Theorem 7.1. *Suppose $\Theta \sim \Lambda$, $X|\Theta = \theta \sim P_\theta$, and $L(\theta, d) \geq 0$ for all $\theta \in \Omega$ and all d . If*

a) $EL(\Theta, \delta_0) < \infty$ for some δ_0 ,

and

b) for a.e. x there exists a value $\delta_\Lambda(x)$ minimizing

$$E[L(\Theta, d) \mid X = x]$$

with respect to d ,

then δ_Λ is a Bayes estimator.

Proof. Let δ be an arbitrary estimator. Then for a.e. x ,

$$\begin{aligned} E[L(\Theta, \delta(X)) \mid X = x] &= E[L(\Theta, \delta(x)) \mid X = x] \\ &\geq E[L(\Theta, \delta_\Lambda(x)) \mid X = x] \\ &= E[L(\Theta, \delta_\Lambda(X)) \mid X = x], \end{aligned}$$

and so

$$E[L(\Theta, \delta(X)) \mid X] \geq E[L(\Theta, \delta_\Lambda(X)) \mid X],$$

almost surely. Taking expectations, by smoothing,

$$EL(\Theta, \delta(X)) \geq EL(\Theta, \delta_\Lambda(X)).$$

Thus δ_Λ is Bayes. □

7.2 Examples

Example 7.2. Weighted Squared Error Loss. Suppose

$$L(\theta, d) = w(\theta)(d - g(\theta))^2.$$

By Theorem 7.1, $\delta_A(x)$ should minimize

$$\begin{aligned} E[w(\Theta)(d - g(\Theta))^2 \mid X = x] &= d^2 E[w(\Theta) \mid X = x] \\ &\quad - 2d E[w(\Theta)g(\Theta) \mid X = x] \\ &\quad + E[w(\Theta)g^2(\Theta) \mid X = x]. \end{aligned}$$

This is a quadratic function of d , minimized when the derivative

$$2dE[w(\Theta) \mid X = x] - 2E[w(\Theta)g(\Theta) \mid X = x]$$

equals zero. Thus

$$\delta_A(x) = \frac{E[w(\Theta)g(\Theta) \mid X = x]}{E[w(\Theta) \mid X = x]}. \quad (7.2)$$

If the weight function w is identically one, then

$$\delta_A(X) = E[g(\Theta) \mid X],$$

called the *posterior mean* of $g(\Theta)$.

If $\mathcal{P}$ is a dominated family with p_θ the density for P_θ , and if Λ is absolutely continuous with Lebesgue density λ , then the joint density of X and Θ is

$$p_\theta(x)\lambda(\theta).$$

By (6.1), the marginal density of X is

$$q(x) = \int p_\theta(x)\lambda(\theta) d\theta,$$

and by (6.4), the conditional density of Θ given $X = x$ is

$$\lambda(\theta|x) = \frac{p_\theta(x)\lambda(\theta)}{q(x)}.$$

Using this, (7.2) becomes

$$\delta_A(x) = \frac{\int w(\theta)g(\theta)p_\theta(x)\lambda(\theta) d\theta}{\int w(\theta)p_\theta(x)\lambda(\theta) d\theta}.$$

(The factor $1/q(x)$ common to both the numerator and denominator cancels.)

Example 7.3. Binomial. When P_θ is the binomial distribution with n trials and success probability θ , the beta distribution $\text{Beta}(\alpha, \beta)$ with density

$$\lambda(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}, \quad \theta \in (0, 1),$$

is a common choice for the prior distribution of Θ . (For a derivation of this density, see Problem 6.1.) The beta density integrates to one, therefore

$$\int_0^1 \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}. \quad (7.3)$$

Using this,

$$\begin{aligned} E\Theta &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{1+\alpha-1} (1 - \theta)^{\beta-1} d\theta \\ &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha + 1)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)} \\ &= \frac{\alpha}{\alpha + \beta}. \end{aligned}$$

The marginal density of X in the Bayesian model is

$$\begin{aligned} q(x) &= \int p_\theta(x) \lambda(\theta) d\theta \\ &= \int_0^1 \binom{n}{x} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{x+\alpha-1} (1 - \theta)^{n-x+\beta-1} d\theta \\ &= \binom{n}{x} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(x + \alpha)\Gamma(n - x + \beta)}{\Gamma(n + \alpha + \beta)}. \end{aligned}$$

This is the mass function for the beta-binomial distribution, sometimes used in a non-Bayesian setting to model variables that exhibit more variation than would be anticipated from a binomial model. Dividing $p_\theta(x)\lambda(\theta)$ by this mass function $q(x)$,

$$\lambda(\theta|x) = \frac{\Gamma(n + \alpha + \beta)}{\Gamma(\alpha + x)\Gamma(\beta + n - x)} \theta^{x+\alpha-1} (1 - \theta)^{n-x+\beta-1}, \quad \theta \in (0, 1).$$

This shows that

$$\Theta|X = x \sim \text{Beta}(x + \alpha, n - x + \beta).$$

The updating necessary to find the posterior distribution from the prior and the observed data is particularly simple here; just increment α by the number of successes observed, and increment β by the number of failures. Prior distributions that ensure a posterior distribution from the same class are called *conjugate*. See Problem 7.4 for a class of examples. With squared error loss,

$$\delta_A(X) = E[\Theta|X] = \frac{X + \alpha}{n + \alpha + \beta}.$$

Straightforward algebra gives

$$\delta_A(X) = \left[\frac{n}{n + \alpha + \beta} \right] \frac{X}{n} + \left[1 - \frac{n}{n + \alpha + \beta} \right] \frac{\alpha}{\alpha + \beta},$$

which shows that the Bayes estimator here is a weighted average of the UMVU estimator X/n and the prior mean $E\Theta = \alpha/(\alpha + \beta)$.

Example 7.4. Negative Binomial. From a sequence of Bernoulli trials with success probability θ , let X be the number of failures before the second success. Then

$$p_\theta(x) = P_\theta(X = x) = (x + 1)\theta^2(1 - \theta)^x, \quad x = 0, 1, \dots$$

Consider estimation of $g(\Theta) = 1/\Theta$ for a Bayesian model in which Θ is uniformly distributed on $(0, 1)$. Then

$$\lambda(\theta|x) \propto_\theta p_\theta(x)\lambda(\theta) \propto_\theta \theta^2(1 - \theta)^x.$$

This is proportional to the density for $\text{Beta}(3, x + 1)$, and so

$$\Theta|X = x \sim \text{Beta}(3, x + 1).$$

The posterior mean is

$$\begin{aligned} \delta_0(x) = E[\Theta^{-1}|X = x] &= \frac{\Gamma(x + 4)}{\Gamma(3)\Gamma(x + 1)} \int_0^1 \theta(1 - \theta)^x d\theta \\ &= \frac{\Gamma(x + 4)\Gamma(2)\Gamma(x + 1)}{\Gamma(3)\Gamma(x + 1)\Gamma(x + 3)} = \frac{x + 3}{2}. \end{aligned}$$

Recalling from Example 5.3 that the UMVU estimator of $1/\theta$ for this model is

$$\delta_1(x) = \frac{x + 2}{2},$$

we have the curious result that

$$\delta_0(X) = \delta_1(X) + \frac{1}{2}.$$

So the estimator δ_0 has constant bias $b(\theta, \delta_0) = E_\theta \delta_0(X) - 1/\theta = 1/2$. With squared error loss, the risk of any estimator is its variance plus the square of its bias. Because δ_0 and δ_1 differ by a constant they have the same variance, and so

$$R(\theta, \delta_0) = \text{Var}_\theta(\delta_0) + 1/4 = \text{Var}_\theta(\delta_1) + 1/4 = R(\theta, \delta_1) + 1/4.$$

Thus the UMVU estimator δ_1 has uniformly smaller risk than δ_0 ! An estimator is called *inadmissible* if a competing estimator has a better¹ risk function. And an inadmissible estimator is generally not Bayes, because an estimator with a better risk function usually has smaller integrated risk. See Theorems 11.6 and 11.7. Trouble arises in this innocuous example because condition (a) in Theorem 7.1 fails. When this happens, any estimator will minimize (7.1), and Bayesian calculations may lead to a poor estimator.

7.3 Utility Theory²

In much of this book there is a basic presumption that risk or expected loss should be used to compare and judge estimators. This may be reasonably intuitive, but there is an important philosophical question regarding why expectation should play such a central role. *Utility theory* provides motivation for this approach, showing that if preferences between probability distributions obey a few basic axioms, then one distribution will be preferred over another if and only if its expected utility is greater. The treatment of utility theory here is a bit sketchy. For more details see Chapter 7 of DeGroot (1970).

Let R be a set of rewards. These rewards could be numerical or monetary, but more ethereal settings in which a reward might be some degree of fame or happiness could also be envisioned. Let $\mathcal{R}$ denote all probability distributions on $(R, \mathcal{F})$, where $\mathcal{F}$ is some σ -field. The distributions $P \in \mathcal{R}$ are called “lotteries.” The idea here is that if you play some lottery $P \in \mathcal{R}$ you will receive a random reward in R according to the distribution P . Let “ $\preceq$ ” indicate preferences among lotteries in $\mathcal{R}$. Formally, $\preceq$ should be a *complete ordering* of $\mathcal{R}$; that is, it should satisfy these conditions:

1. If P_1 and P_2 are lotteries in $\mathcal{R}$, then either $P_1 \prec P_2$, $P_2 \prec P_1$, or $P_1 \simeq P_2$. (Here $P_1 \simeq P_2$ means that $P_1 \preceq P_2$ and $P_2 \preceq P_1$; and $P_1 \prec P_2$ means that $P_1 \preceq P_2$, but $P_1 \not\simeq P_2$.)
2. If P_1 , P_2 , and P_3 are lotteries in $\mathcal{R}$ with $P_1 \preceq P_2$ and $P_2 \preceq P_3$, then $P_1 \preceq P_3$.

It is also convenient to identify a reward $r \in R$ with the degenerate probability distribution in $\mathcal{R}$ that assigns unit mass to $\{r\}$. (To ensure this is possible, the σ -field $\mathcal{F}$ must contain all singletons $\{r\}$, $r \in R$.) We can then define reward intervals

$$[r_1, r_2] = \{r \in R : r_1 \preceq r \preceq r_2\}.$$

A lottery $P \in \mathcal{R}$ is called *bounded* if $P([r_1, r_2]) = 1$ for some rewards r_1 and r_2 in R . Let $\mathcal{R}_B$ denote the collection of all bounded lotteries in $\mathcal{R}$.

¹ See Section 11.3 for a formal definition.

² This section covers optional material not used in later chapters.

Definition 7.5. A (measurable) function $U : R \rightarrow \mathbb{R}$ is a utility function for $\precsim$ if

$$P_1 \precsim P_2 \text{ if and only if } E_{P_1} U \leq E_{P_2} U,$$

whenever the expectations exist. (Here $E_P U = \int U dP$.)

The following example shows that utility functions may or may not exist.

Example 7.6. Suppose R contains two rewards, a and b , and let P_θ be the lottery that gives reward a with probability θ . Suppose $b \prec a$. Then if $\precsim$ has a utility function U , $U(a)$ is larger than $U(b)$. Inasmuch as the expected utility of P_θ is

$$\int U(r) dP_\theta(r) = \theta U(a) + (1 - \theta)U(b) = \theta[U(a) - U(b)] + U(b),$$

the expected utility of P_θ increases as θ increases. Hence $P_{\theta_1} \precsim P_{\theta_2}$ if and only if $\theta_1 \leq \theta_2$. Similarly, if $a \simeq b$, then $U(a) = U(b)$ and all lotteries are equivalent under $\precsim$. But preferences between lotteries do not have to behave in this fashion. For instance, if someone views rewards a and b as comparable, but finds pleasure in the excitement of not knowing the reward they will receive, a preference relation in which $P_{\theta_1} \precsim P_{\theta_2}$ if and only if $|\theta_1 - 1/2| \geq |\theta_2 - 1/2|$ may be appropriate. For this preference relation there is no utility function.

Under axioms given below, utility functions will exist. The language makes extensive use of pairwise mixtures of distributions. If P_1 and P_2 are lotteries and $\alpha \in (0, 1)$, then the mixture $\alpha P_1 + (1 - \alpha)P_2$ can be viewed (by smoothing) as a lottery that draws from P_1 with probability α and draws from P_2 with probability $1 - \alpha$. In particular, because we associate rewards with degenerate lotteries, $\alpha r_1 + (1 - \alpha)r_2$ represents a lottery that gives reward r_1 with probability α and reward r_2 with probability $1 - \alpha$.

A1) If P_1 , P_2 , and P are bounded lotteries in $\mathcal{R}_B$, and if $\alpha \in (0, 1)$, then $P_1 \prec P_2$ if and only if $\alpha P_1 + (1 - \alpha)P \prec \alpha P_2 + (1 - \alpha)P$.

It is also easy to see that $P_1 \precsim P_2$ if and only if $\alpha P_1 + (1 - \alpha)P \precsim \alpha P_2 + (1 - \alpha)P$. As a further consequence, if $P_1 \precsim Q_1$ and $P_2 \precsim Q_2$, all in $\mathcal{R}_B$, and $\alpha \in (0, 1)$, then

$$\alpha P_1 + (1 - \alpha)P_2 \precsim \alpha Q_1 + (1 - \alpha)P_2 \precsim \alpha Q_1 + (1 - \alpha)Q_2.$$

If $P_1 \simeq Q_1$ and $P_2 \simeq Q_2$, again all in $\mathcal{R}_B$, the reverse inequalities also hold, and

$$\alpha P_1 + (1 - \alpha)P_2 \simeq \alpha Q_1 + (1 - \alpha)Q_2. \quad (7.4)$$

As a final consequence of this axiom, if $r_1 \prec r_2$ are rewards in R , and if α and β are constants in $[0, 1]$, then

$$\alpha r_2 + (1 - \alpha)r_1 \prec \beta r_2 + (1 - \beta)r_1 \text{ if and only if } \alpha < \beta. \quad (7.5)$$

A2) If $P_1 \prec P \prec P_2$ are bounded lotteries, then there exist constants α and β in $(0, 1)$ such that $P \prec \alpha P_2 + (1 - \alpha)P_1$ and $P \succ \beta P_2 + (1 - \beta)P_1$.

The following result follows from this axiom, and is used shortly to construct a candidate utility function.

Theorem 7.7. *If $r_1 \precsim r \precsim r_2$ are rewards in R , then there exists a unique value $\nu \in [0, 1]$ such that*

$$r \simeq \nu r_2 + (1 - \nu)r_1.$$

Proof. Consider $S = \{\alpha \in [0, 1] : r \prec \alpha r_2 + (1 - \alpha)r_1\}$, an interval by (7.5), and let ν be the lower endpoint of S , $\nu = \inf S$. If $\nu r_2 + (1 - \nu)r_1 \prec r$ then $\nu < 1$, and by the second axiom

$$r \succ \beta r_2 + (1 - \beta)(\nu r_2 + (1 - \nu)r_1) = (\nu + \beta(1 - \nu))r_2 + (1 - \nu - \beta(1 - \nu))r_1$$

for some $\beta \in (0, 1)$. This would imply that ν is not the lower endpoint of S . But if $\nu r_2 + (1 - \nu)r_1 \succ r$, then $\nu > 0$, and by the second axiom

$$r \prec \alpha r_1 + (1 - \alpha)(\nu r_2 + (1 - \nu)r_1) = (1 - \alpha)\nu r_2 + (1 - (1 - \alpha)\nu)r_1,$$

for some $\alpha \in (0, 1)$, again contradicting $\nu = \inf S$. Thus $r \simeq \nu r_2 + (1 - \nu)r_1$. Uniqueness follows from similar considerations. $\square$

Let $s_0 \prec s_1$ be fixed rewards in R . Utility functions, if they exist, are not unique, for if U is a utility, and if a and b are constants with $b > 0$, then $a + bU$ is also a utility function. From this, if a utility function exists, there will be a utility function with $U(s_0) = 0$ and $U(s_1) = 1$. The construction below gives this utility function.

Suppose $r \in [s_0, s_1]$. Then by Theorem 7.7,

$$r \simeq \nu s_1 + (1 - \nu)s_0,$$

for some $\nu \in [0, 1]$. If a utility function exists, then the expected utilities for the two lotteries in this equation must agree, which means that we must have

$$U(r) = \nu.$$

If instead $r \precsim s_0$, then by Theorem 7.7,

$$s_0 \simeq \nu s_1 + (1 - \nu)r,$$

for some $\nu \in (0, 1)$. Equating expected utilities, $0 = \nu + (1 - \nu)U(r)$, and so we need

$$U(r) = -\frac{\nu}{1 - \nu}.$$

Finally, if $s_1 \prec r$, then by Theorem 7.7,

$$s_1 \simeq \nu r + (1 - \nu)s_0,$$

and equating expected utilities we must have

$$U(r) = \frac{1}{\nu}.$$

The following technical axiom is needed to ensure that this function U is measurable.

A3) For any r_1, r_2 , and r_3 in R , and any α and β in $[0, 1]$,

$$\{r \in R : \alpha r + (1 - \alpha)r_1 \precsim \beta r_2 + (1 - \beta)r_3\} \in \mathcal{F}.$$

Let P be a bounded lottery, so that $P\{[r_1, r_2]\} = 1$ for some r_1 and r_2 in R . The final axiom concerns a two-stage lottery in which the first stage is P , and the second stage trades in P for an equivalent mixture of r_1 and r_2 . To be specific, define a function $\alpha : [r_1, r_2] \rightarrow [0, 1]$ using Theorem 7.7 so that

$$r \simeq \alpha(r)r_2 + (1 - \alpha(r))r_1.$$

From the construction of U it can be shown that

$$\alpha(r) = \frac{U(r) - U(r_1)}{U(r_2) - U(r_1)}. \quad (7.6)$$

For instance, if $s_0 \precsim r_1 \precsim r \precsim r_2 \precsim s_1$, from the construction of U ,

$$r_1 \simeq U(r_1)s_1 + (1 - U(r_1))s_0, \quad r_2 \simeq U(r_2)s_1 + (1 - U(r_2))s_0,$$

and, using (7.4),

$$\begin{aligned} \alpha r_2 + (1 - \alpha)r_1 &\simeq \alpha[U(r_2)s_1 + (1 - U(r_2))s_0] + (1 - \alpha)[U(r_1)s_1 + (1 - U(r_1))s_0] \\ &= [\alpha U(r_2) + (1 - \alpha)U(r_1)]s_1 + [1 - \alpha U(r_2) - (1 - \alpha)U(r_1)]s_0. \end{aligned}$$

Because $r \simeq U(r)s_1 + (1 - U(r))s_0$, $r \simeq \alpha r_2 + (1 - \alpha)r_1$ when

$$U(r) = \alpha U(r_2) + (1 - \alpha)U(r_1).$$

Solving for α we obtain (7.6).

In the two-stage lottery, if the reward for the first stage, sampled from P , is r , then the second stage is $\alpha(r)r_2 + (1 - \alpha(r))r_1$. Conditioning on the outcome of the first stage, this two-stage lottery gives reward r_2 with probability

$$\beta = \int \alpha(r) dP(r).$$

Otherwise, the two-stage lottery gives reward r_1 . The final axiom asserts that under $\precsim$ this two-stage lottery is equivalent to P .

A4) $P \simeq \beta r_2 + (1 - \beta)r_1$.

Based on the stated axioms, the final result of this section shows that the function U constructed above is a utility function for bounded lotteries.

Theorem 7.8. *If axioms A1 through A4 hold, and P_1 and P_2 are bounded lotteries, then*

$$P_1 \precsim P_2 \text{ if and only if } E_{P_1}U \leq E_{P_2}U.$$

Proof. Choose r_1 and r_2 so that $P_1\{[r_1, r_2]\}$ and $P_2\{[r_1, r_2]\}$ both equal one. By the fourth axiom and (7.6),

$$P_1 \simeq \left[\frac{E_{P_1}U - U(r_1)}{U(r_2) - U(r_1)} \right] r_2 + \left[\frac{U(r_2) - E_{P_1}U}{U(r_2) - U(r_1)} \right] r_1$$

and

$$P_2 \simeq \left[\frac{E_{P_2}U - U(r_1)}{U(r_2) - U(r_1)} \right] r_2 + \left[\frac{U(r_2) - E_{P_2}U}{U(r_2) - U(r_1)} \right] r_1.$$

By (7.5), $P_1 \precsim P_2$ if and only if

$$\frac{E_{P_1}U - U(r_1)}{U(r_2) - U(r_1)} \leq \frac{E_{P_2}U - U(r_1)}{U(r_2) - U(r_1)},$$

which happens if and only if $E_{P_1}U \leq E_{P_2}U$. □

7.4 Problems³

- *1. Consider a Bayesian model in which the prior distribution for Θ is exponential with failure rate η , so that $\lambda(\theta) = \eta e^{-\eta\theta}$, $\theta > 0$. Given $\Theta = \theta$, the data $X_1, \dots, X_n$ are i.i.d. from the Poisson distribution with mean θ . Determine the Bayes estimator for Θ if the loss function is $L(\theta, d) = \theta^p(d - \theta)^2$, with p a fixed positive constant.
- *2. Consider a Bayesian model in which the prior distribution for Θ is absolutely continuous with density $\lambda(\theta) = 1/(1 + \theta)^2$, $\theta > 0$. Given $\Theta = \theta$, our datum is a single variable X uniformly distributed on $(0, \theta)$. Give an equation to find the Bayes estimate $\delta_A(X)$ of Θ if the loss function is $L(\theta, d) = |d - \theta|$. Determine $P(\delta_A(X) < \Theta | X = x)$, explicitly.
- *3. In a Bayesian approach to simple linear regression, suppose the intercept Θ_1 and slope Θ_2 of the regression line are *a priori* independent with $\Theta_1 \sim N(0, \tau_1^2)$ and $\Theta_2 \sim N(0, \tau_2^2)$. Given $\Theta_1 = \theta_1$ and $\Theta_2 = \theta_2$, data $Y_1, \dots, Y_n$ are independent with $Y_i \sim N(\theta_1 + \theta_2 x_i, \sigma^2)$, where the variance σ^2 is known, and $x_1, \dots, x_n$ are constants summing to zero, $x_1 + \dots + x_n = 0$. Find the Bayes estimates of Θ_1 and Θ_2 under squared error loss.

³ Solutions to the starred problems are given at the back of the book.

- *4. *Conjugate prior distributions.* Let $\mathcal{P} = \{P_\theta, \theta \in \Omega\}$ be a one-parameter canonical exponential family with densities p_θ given by

$$p_\theta(x) = h(x)e^{\theta T(x) - A(\theta)}.$$

Here Ω is an interval. Let $\Lambda = \Lambda_{\alpha, \beta}$ be an absolutely continuous prior distribution with density

$$\lambda(\theta) = \begin{cases} \exp\{\alpha\theta - \beta A(\theta) - B(\alpha, \beta)\}, & \theta \in \Omega; \\ 0, & \text{otherwise,} \end{cases}$$

where

$$B(\alpha, \beta) = \log \int_{\Omega} \exp\{\alpha\theta - \beta A(\theta)\} d\theta.$$

These densities $\Lambda_{\alpha, \beta}$ form a canonical two-parameter exponential family. Let $\Xi = \{(\alpha, \beta) : B(\alpha, \beta) < \infty\}$ be the canonical parameter space. Assume for regularity that $\lambda(\theta) \rightarrow 0$ as θ approaches either end of the interval Ω , regardless the value of $(\alpha, \beta) \in \Xi$.

- a) With the stated regularity, $\int_{\Omega} \lambda'(\theta) d\theta = 0$. Use this to give an explicit formula for $EA'(\theta)$ when $\Theta \sim \Lambda_{\alpha, \beta}$. (The answer should be a simple function of α and β .)
 - b) Consider a Bayesian model in which $\Theta \sim \Lambda_{\alpha, \beta}$ and given $\Theta = \theta$, $X_1, \dots, X_n$ are i.i.d. with common density P_θ from the exponential family $\mathcal{P}$. Determine the Bayes estimate of $A'(\theta)$ under squared error loss. Show that this estimate is a weighted average of $EA'(\theta)$ and the average $\bar{T} = [T(X_1) + \dots + T(X_n)]/n$.
 - c) Demonstrate the ideas in parts (a) and (b) when P_θ is the exponential distribution with failure rate θ and mean $1/\theta$: identify the prior distributions $\Lambda_{\alpha, \beta}$, and give an explicit formula for the Bayes estimator of the mean $1/\theta$.
5. Consider an autoregressive model in which $X_1 \sim N(\theta, \sigma^2/(1 - \rho^2))$ and the conditional distribution of X_{j+1} given $X_1 = x_1, \dots, X_j = x_j$ is $N(\theta + \rho(x_j - \theta), \sigma^2)$, $j = 1, \dots, n-1$. Suppose the values for ρ and σ are fixed constants, and consider Bayesian estimation with $\Theta \sim N(0, \tau^2)$. Find Bayes estimates for Θ and Θ^2 under squared error loss.
- *6. Consider a Bayesian model in which the random parameter Θ has a Bernoulli prior distribution with success probability $1/2$, so $P(\Theta = 0) = P(\Theta = 1) = 1/2$. Given $\Theta = 0$, data X has density f_0 , and given $\Theta = 1$, X has density f_1 .
- a) Find the Bayes estimate of Θ under squared error loss.
 - b) Find the Bayes estimate of Θ if $L(\theta, d) = I\{\theta \neq d\}$ (called *zero-one loss*).
- *7. Consider Bayesian estimation in which the parameter Θ has a standard exponential distribution, so $\lambda(\theta) = e^{-\theta}$, $\theta > 0$, and given $\Theta = \theta$, $X_1, \dots, X_n$ are i.i.d. from an exponential distribution with failure rate θ . Determine the Bayes estimator of Θ if the loss function is $L(\theta, d) = (d - \theta)^2/d$.

8. Consider a Bayesian model in which the prior distribution for Θ is standard exponential and the density for X given $\Theta = \theta$ is

$$p_{\theta}(x) = \begin{cases} e^{\theta-x}, & x > \theta; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the marginal density for X and EX in the Bayesian model.
 b) Find the Bayes estimator for Θ under squared error loss. (Assume $X > 0$.)
9. Suppose $\Theta \sim \Lambda$ and $X|\Theta = \theta \sim P_{\theta}$, and let f be a nonnegative measurable function. Use smoothing to write $Ef(\Theta, X)$ as an iterated integral. (This calculation shows that specification of a Bayesian model in this fashion determines the joint distribution of X and Θ .)
10. Suppose we observe two independent observations, (X_1, Y_1) and (X_2, Y_2) from an absolutely continuous bivariate distribution with density

$$\frac{\sqrt{1-\theta^2}}{2\pi} \exp\left[-\frac{1}{2}(x^2 + y^2 - 2\theta xy)\right].$$

Find the Bayes estimate for Θ under squared error loss if the prior distribution is uniform on $(-1, 1)$.

11. Consider a Bayesian model in which the prior distribution for Θ is uniform on $(0, 1)$ and given $\Theta = \theta$, X_i , $i \geq 1$, are i.i.d. Bernoulli with success probability θ . Find

$$P(X_{n+1} = 1 | X_1, \dots, X_n).$$

12. *Bayesian prediction.*

- a) Let X and Y be jointly distributed, with X a random variable and Y a random vector. Suppose we are interested in predicting X from Y . The efficacy of a predictor $f(Y)$ might be measured using the expected squared error, $E(X - f(Y))^2$. Use a smoothing argument to find the function f minimizing this quantity.
- b) Consider a Bayesian model in which Θ is a random parameter, and given $\Theta = \theta$, random variables $X_1, \dots, X_{n+1}$ are i.i.d. from a distribution P_{θ} with mean $\mu(\theta)$. With squared error loss, the best estimator of $\mu(\theta)$ based on $X_1, \dots, X_n$ is

$$\hat{\mu} = E[\mu(\Theta) | X_1, \dots, X_n].$$

Show that $\hat{\mu}$ is also the best predictor for X_{n+1} based on $Y = (X_1, \dots, X_n)$. You can assume that Θ is absolutely continuous, and that the family $\mathcal{P} = \{P_{\theta} : \theta \in \Omega\}$ is dominated with densities p_{θ} , $\theta \in \Omega$.

13. Consider a Bayesian model in which Θ is absolutely continuous with density

$$\lambda(\theta) = \begin{cases} \frac{e^{-1/\theta}}{\theta^2}, & \theta > 0; \\ 0, & \text{otherwise,} \end{cases}$$

and given $\Theta = \theta$, $X_1, \dots, X_n$ are i.i.d. $N(0, \theta)$. Find the Bayes estimator for Θ under squared error loss.

14. Consider a Bayesian model in which given $\Theta = \theta$, $X_1, \dots, X_n$ are i.i.d. from a Bernoulli distribution with mean θ .

a) Let $(\pi(1), \dots, \pi(n))$ be a permutation of $(1, \dots, n)$. Show that

$$(X_{\pi(1)}, \dots, X_{\pi(n)}) \quad \text{and} \quad (X_1, \dots, X_n)$$

have the same distribution. When this holds the variables involved are said to be *exchangeable*.

b) Show that $\text{Cov}(X_i, X_j) \geq 0$. When will this covariance be zero?

15. Consider a Bayesian model in which Θ is absolutely continuous with density

$$\lambda(\theta) = \begin{cases} \frac{4\theta^3}{(1+\theta)^5}, & \theta > 0; \\ 0, & \text{otherwise,} \end{cases}$$

and given $\Theta = \theta > 0$, data X and Y are absolutely continuous with density

$$p_\theta(x, y) = \begin{cases} 1/\theta^3, & |x| < \theta y < \theta^2; \\ 0, & \text{otherwise,} \end{cases}$$

Find the Bayes estimator of Θ under squared error loss.

- *16. (For fun) Let X and Y be independent Cauchy variables with location θ .
- a) Show that X and the average $A = (X + Y)/2$ have the same distribution.
- b) Show that $P_\theta(|A - \theta| < |X - \theta|) > 1/2$, so that A is more likely to be closer to θ than X . (Hint: Graph the region in the plane where the event in question occurs.)

